

Xiaochang(Shaw) Li

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EDUCATION

Carnegie Mellon University

Master of Science in Information Technology – Mobility

Pittsburgh, PA & San Jose, CA

August 2025 - January 2027

- GPA: 4.0/4.0
- Coursework: Introduction to Computer Systems, Cloud Infrastructure and Services, Database Systems

Fudan University

Bachelor of Science in Computer Science

Shanghai, China

September 2020 - August 2024

- GPA: 90/100
- Coursework: Data Structures & Algorithms, Computer Systems, Computer Networks, Parallel Distributed Computing
- Honors & Reward: Scholarship of Fudan University, Fudan Outstanding Student

WORK EXPERIENCE

Tiktok

San Jose, CA

SOFTWARE ENGINEERING INTERN

May 2026 - Present

- Designed a replayable, self-improving diagnosis platform that routes requests through deterministic expert rules with a ReAct fallback, continuously compiling validated agent reasoning into production rules through offline learning
- Built replayable snapshots by normalizing 8+ heterogeneous infrastructure data sources into a shared intermediate representation with provenance-aware lineage, enabling deterministic replay, cross-workflow evidence reuse, and reducing upstream IO by 66%
- Designed an extensible execution model that separates data acquisition, normalization, and diagnosis into pluggable workflow nodes, allowing new infrastructure systems to integrate without modifying existing diagnosis logic
- Engineered a human-gated continuous learning pipeline where agent-diagnosed misses automatically generate candidate rule, undergo AI-assisted review and full-corpus replay validation before approval, promoting 500+ production rules into deterministic execution

Bytedance

Shanghai, China

SOFTWARE ENGINEERING INTERN

December 2024 - April 2025

- Identified bottlenecks in test data/case management; developed CI/CD workflows automating build, test (Pytest), and release, cut end-to-end duration by 20% and saving 250+ annual hours
- Spearheaded LLM-based DOM analysis framework generating 85% of functional test cases; authored 200+ backend Pytest suites, collectively increasing overall code coverage from 70% to 80%
- Collaborated with Product/UI to co-author 50+ API specs for Stream Resource features, ensuring clear contracts and cutting down integration friction

Xiaoyong Private Equity Funds

Shanghai, China

SOFTWARE ENGINEER INTERN

June 2023 - September 2023

- Optimized IPC for high-frequency market data feeds by replacing UDS with DPDK/libmemif zero-copy channels, attaining 23% lower latency (2.4μs→1.84μs in bulk data transfers under localhost benchmark)
- Refactored messaging system from scratch to custom C++ pub-sub model, decoupling producers/consumers and enhancing flexibility/scalability
- Authored 80+ Google Test suites for IPC libraries, achieving 85% coverage and bolstering robustness

KEY PROJECTS

High-Performance Dynamic Memory Allocator | C, Systems Programming

- Implemented segregated free lists with 12 size classes and mini-block optimization for 8-16B allocations, achieving 74% memory utilization and 9,662 Kops/sec throughput (100/100 performance score)
- Eliminated footers for allocated blocks by encoding previous allocated status in header bits, reducing metadata overhead by 50% while maintaining O(1) coalescing through bi-directional list traversal
- Optimized for SPEC benchmarks through first-fit policy and immediate coalescing, outperforming baseline (7,242 Kops/sec) by 33% in all workloads

COOL Compiler | C++, Flex, Bison

- Designed an end-to-end compiler for COOL, covering lexical analysis, parsing, semantic validation, and MIPS code generation
- Built a stateful Flex scanner and LALR(1) Bison parser supporting nested comments, string-error handling, operator precedence, error recovery, and abstract syntax tree construction.
- Implemented inheritance-graph validation, cycle detection, scoped symbol resolution, SELF_TYPE checking, and lowering from typed program representations to executable MIPS assembly.

SKILLS

Programming Languages: C/C++, Python, Golang, SQL, TypeScript, Terraform, Shell, Java

Systems & Data: Linux, RPC, Raft, MapReduce, Spark, YARN, Kafka, DPDK/libmemif, MySQL

Cloud & Infrastructure: Kubernetes, Docker, Terraform, GCP/GKE, AWS, Jenkins, CI/CD

Testing & Reliability: Fault diagnosis, snapshot replay, data lineage, performance profiling, Pytest, Google Test, SonarQube